De La Salle University-Dasmariñas

ROAD DAMAGE DETECTION WEB APPLICATION FOR LGU ROAD SURVEYS USING YOLOv8

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CHAPTER 1

INTRODUCTION

1.1 Background of the Study

1.1.1 Project Context

Potholes are among the most common and hazardous forms of road damage, posing a persistent challenge for local government units (LGUs) tasked with maintaining infrastructure quality. Potholes lead to vehicular accidents, increased vehicle maintenance costs, and disrupted transportation networks. In the Philippines, road inspections are still heavily conducted manually. Personnel must travel along road segments to visually identify and manually record findings. This process is slow, labor-intensive, and subjective, often resulting in unreported damage that worsens over time.

Recent advancements in computer vision, particularly the You Only Look Once (YOLO) architecture introduced by Redmon et al. (2016), have proven highly effective in automating road damage detection due to their ability to process frames rapidly. The availability of large-scale datasets, such as the RDD2022 dataset containing over 47,000 annotated road images across multiple countries (Arya et al., 2022), has allowed researchers to train highly accurate detection models. Successive improvements in YOLO architectures have demonstrated exceptional speed-accuracy trade-offs for detecting road anomalies, making them ideal for processing continuous video feeds (Pham et al., 2024). Further studies by Pham et al. (2022), and Roy and Bhaduri (2023) have routinely validated the high precision of YOLO models on road damage detection.

Despite these global advances, solutions tailored for the Philippine context remain limited in their deployment and practicality. Sarmiento (2021) validated YOLOv4 on a small set of Philippine pavement images but only as a research experiment with no deployable system. Fortin et al. (2024) developed the EyeWay system, which mapped potholes, but required dedicated hardware devices that add costs smaller LGUs cannot afford. Similarly, Bustria et al. (2025) developed a YOLOv11-based mobile app for Baguio City; however, their system relies on users uploading individual static images for classification rather than processing continuous video to pinpoint damage locations automatically. Instead of relying on live mobile-app processing, which is limited by smartphone hardware constraints in the field , this study utilizes a centralized web application. While newer versions like YOLOv11 exist, YOLOv8 is utilized in this research to provide a stable and optimized architecture for processing frame-by-frame video data within a web environment.

Instead of relying on live mobile-app processing, which is limited by smartphone hardware constraints in the field , a more practical workflow is a centralized web application. While newer iterations like YOLOv11 have been explored in recent local studies , this research utilizes YOLOv8 for its established stability and efficient frame-by-frame processing within a web environment. By capturing road conditions via phone cameras or car dashcams during routine patrols and subsequently processing that video footage through a web app, LGUs can efficiently decouple fieldwork data collection from heavy computational analysis.

1.1.2 Purpose and Description

This study aims to develop a web-based application for LGUs designed to automate the detection and localization of potholes from uploaded road survey videos. Rather than requiring real-time, on-device mobile hardware, authorized LGU personnel can upload continuous video footage to the web platform.

The system will utilize a trained YOLO object detection model (e.g., YOLOv8) to process the video frame-by-frame. Upon detecting a pothole, the web application will visually highlight the damage using bounding boxes directly within the video frame and extract the exact timestamp of the detection. This provides engineers and road maintenance teams with an automated, searchable log of road damage, allowing them to instantly navigate to the specific timestamp of the video to view the pothole's severity and location context.

1.2 Statement of the Problem

Existing road inspection and assessment methods are limited by their reliance on manual, on-site visual surveys and static image-based applications. These approaches frequently fail to leverage continuous video data, which is essential for safely and efficiently mapping expansive road networks. As a result, road damage documentation often lacks precise spatial and temporal context, leading to inefficient LGU resource allocation, delayed repairs, and limited decision-support.

This study addresses the following research questions:

1. How can object detection models (e.g., YOLOv8) be utilized to effectively detect and localize potholes from uploaded road survey videos?
2. In what way can a web application integrate pothole detection data with contextual video information (specifically, exact timestamps and visual bounding boxes) to generate an actionable damage log?
3. How does incorporating automated video analysis and timestamp generation improve road inspection efficiency for LGUs compared to traditional manual inspection methods?
4. What limitations and challenges arise in developing a video-based pothole detection web application, particularly in terms of video quality variations, processing speed, and model, processing constraints, and detection accuracy?

1.3 Research Objectives

1.3.1 General Objective

To design, develop, and evaluate a web application that automatically analyzes uploaded road survey videos to detect potholes, highlight their locations within the video frames, and generate precise timestamps for each detection.

1.3.2 Specific Objectives

1. To develop a web-based user interface that allows LGU personnel to easily upload and manage road survey videos.

2. To train and integrate a YOLOv8-based object detection model optimized for processing video frames to detect potholes accurately using datasets like RDD2022 and locally collected images.
3. To implement a frame-analysis feature that draws bounding boxes over detected potholes directly on the video feed.
4. To extract the exact timestamp of every detected pothole and generate a clickable log so users can quickly review the relevant part of the video.
5. To evaluate the model's detection performance in terms of mean Average Precision (mAP), Precision, and Recall.

1.4 Significance of the Study

The development of this web-based pothole detection system provides meaningful benefits to various stakeholders by addressing the inefficiencies of manual road inspection.

- **Local Government Units (LGUs):** Provides an automated tool to replace slow, labor-intensive manual recording processes, allowing for more frequent and comprehensive infrastructure monitoring.
- **Road Maintenance Teams and Engineers:** Offers an actionable, searchable log of road damage with precise timestamps. This enables teams to instantly navigate to specific video segments to assess damage severity and prioritize repairs.

- **The General Public:** By facilitating faster repairs and better resource allocation, the study contributes to reduced vehicular accidents and lower vehicle maintenance costs for citizens.
- **Future Researchers:** Serves as a practical implementation of YOLO-based object detection tailored for the Philippine context. It provides a baseline for decoupling fieldwork data collection from heavy computational analysis using web-based workflows.

1.5 Scope of the Study

This study focuses on the creation of a centralized web application designed to automate road damage identification for LGUs. The boundaries of the research are defined as follows:

- **Targeted Damage Type:** The system is exclusively designed to detect and localize potholes. Other forms of road damage (such as longitudinal or alligator cracks) are excluded from the scope because their markings are often inconsistent and vary heavily, making them less suitable for the current model's focus.
- **Platform and Deployment**: The study will develop a web-based application rather than a real-time mobile app. Users will capture road conditions via mobile cameras or dashcams and subsequently upload the continuous video footage to the web platform for processing.
- **Technical Implementation:** The system will utilize the YOLOv8 object detection architecture. The application will process uploaded videos

frame-by-frame to draw bounding boxes around detected potholes and generate a clickable log based on the exact detection timestamps.

- **Data Sources:** The model will be trained using the RDD2022 dataset, which contains annotated road images, supplemented by locally collected images to ensure relevance to Philippine road conditions.
- **Limitations:** The system's performance is subject to variations in video quality and processing constraints. Furthermore, the analysis is performed asynchronously on the web server after the video upload is complete.

CHAPTER 2

REVIEW OF RELATED LITERATURE

CHAPTER 3

METHODOLOGY

3.1 Research Design

3.2 System Development Method

3.2.1 Planning

3.2.2 Analysis

3.2.3 Design

3.2.4 Implementation

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3.3 Data Collection and Dataset Preparation

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3.3.2 Text Preprocessing

3.3.3 Data for Recommendations

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3.4.2 SBERT (Deep Learning - Semantic Embedding Model)

3.4.3 KNN (Machine Learning - Cosine Similarity Recommendation)

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REFERENCES